Practice Problem 3 Guide Questions

1. The space before %c tells scanf() to skip any whitespace left in the input stream before reading the character. This prevents a leftover newline from being read as the user's response. This pattern is commonly used when reading character input after a previous scanf() call.

2. Write it as cost += 25. Look at the sample solution for reference. Whenever we have this accumulator pattern (i.e., give the variable a new value based on its old value plus X), it is assumed that this variable was initialized. If not, we might be dealing with a garbage value.

3. Refer to the implementation. I used resp which is a reusable variable to hold the response of the user for Y/N questions. It is appropriate because we don't need to remember the response of the user after having performed the required action. Except only for the response to the weekend question (see next question).

4. Because we are using a reusable variable, we need to remember the response of the user for the weekend question because it will be used for later. The order of the questions is specific so we cannot rearrange anything. One technique is to use a flag variable. A typical pattern involves initializing this flag to false (in this case) and turn it on if the condition is met. Please refer to the sample code.

5. Refer to the different patterns and point them out. The ticket price uses an \texttt{if-else if-else} ladder because the guest belongs to exactly one age group. The weekend or weekday fee uses an \texttt{if-else} because the visit can only be one or the other. The optional add-ons use separate \texttt{if} statements because the guest may select neither, one, or both add-ons, so the choices are independent. The VIP upgrade uses a nested \texttt{if} because the program should ask about the upgrade only if the guest is at least 18 years old. Different conditional structures are appropriate because each situation has a different relationship among its conditions, such as mutually exclusive choices, independent choices, or decisions that depend on another condition.